

Example: First Principles Driven Hybrid Model for Heat Exchanger Fouling calculations

Problem Statement

How to develop a First Principles Driven Hybrid Model for Heat Exchanger Fouling?

Solution

Heat Exchanger Fouling can change with time and process conditions. Building a customized model can be challenging and often requires significant time and effort, and additional costs.

One solution is to build a predictive heat exchanger model and incorporate historical data from the heat exchanger to capture the fouling factors from the process using a First Principles Driven Hybrid Model.

In this article, we highlight the different steps required to create such a First Principles Driven Hybrid Model to accurately capture these effects.

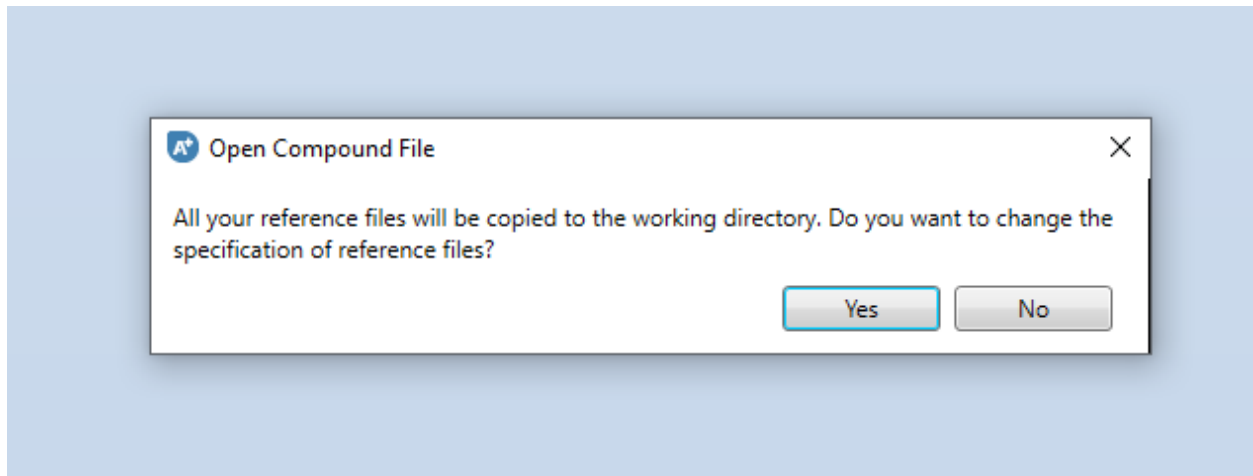
- Step 1 – Import Data
- Step 2 – Condition Data
- Step 3 – Evaluate Model
- Step 4 – Build, Train and Validate Model
- Step 5 – Deploy Model

General Workflow

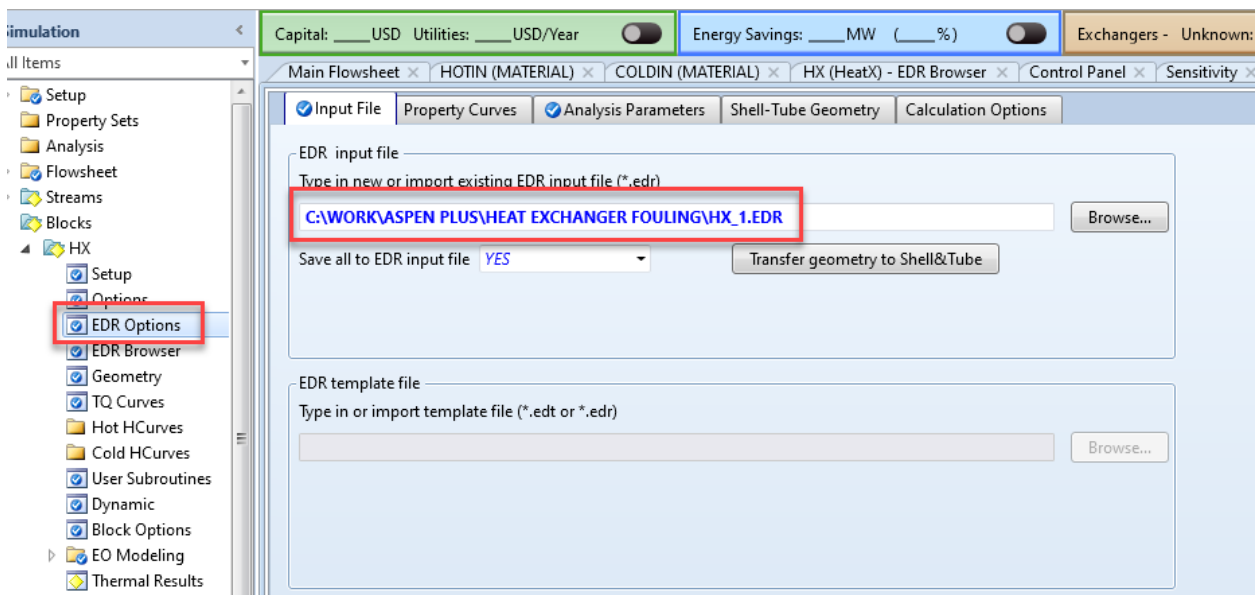


Step 1 – Import Data

1. Open HXFouling_Start.apwz. A message regarding reference files will pop up. Simply click 'No'. We will manually set it up in the next step.



2. Go to the HX Block folder in the Navigation Pane and select EDR Options. Ensure that the EDR input file points to the EDR file HX_1.EDR provided separately in the attachments.



Note: Please do not skip this step. Please make sure it is highlighting the EDR file provided separately, or you may experience issues when you are building the FP Driven Hybrid Model.

3. Run the Aspen Plus simulation to ensure the exchanger solves and Results are available.
4. On the Home ribbon, click the AI Training button.

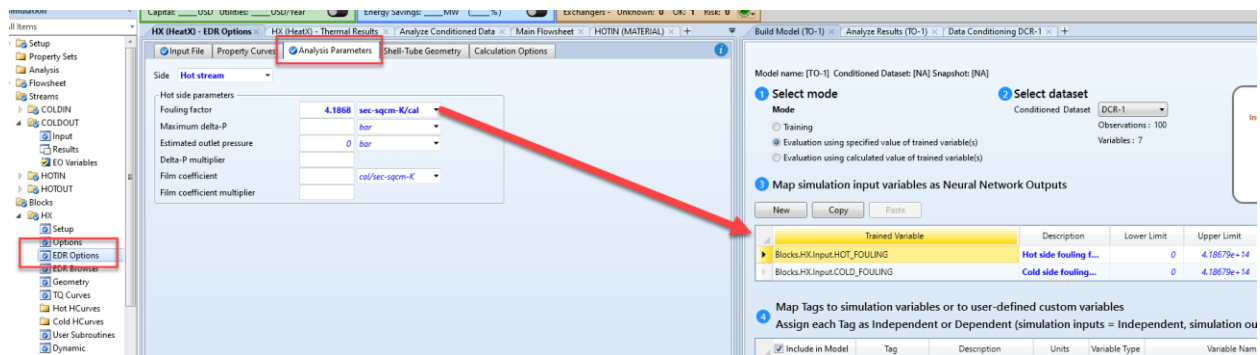
5. Enter ID: TO-1 and click OK.
6. Click on Get Plant Data | Import from Excel.
7. Select HXFouling_PlantData.xlsm.
8. Click >> to add all the available datasets and click Import. After Plant Data has been successfully imported, close the window.
7. Map the units as required and select the Update check box, and then click Apply.
8. Click Done to close the UOM Successfully updated dialog box.
9. Feel free to analyze the raw data - Observe different plots available for various HX variables and review the correlation matrix.

Step 2 – Condition Data

10. We have previously removed all outliers from our dataset, so we will leave all the conditioning settings to their defaults and click on Condition Dataset and start the conditioning for the entire interval.
11. After conditioning is successfully completed, click Done. Dataset DCR-1 has been created.
12. The Analyze Conditioned Data form automatically appears. This form helps you compare between conditioned and raw data.
13. Move to Full Tables. The All Data and Summary Statistics tables can help you visualize your conditioned data.

Step 3 – Evaluate Model

14. From the ribbon, click Build Model and complete the mapping.
15. We want to check if the current model represents the data adequately and if there is a need to build a hybrid model. Select the mode: Evaluation using specified value of trained variable(s)
16. Select the dataset: DCR-1
17. Map simulation input variables as Neural Network Outputs which will be the heat exchanger fouling factors.



18. Map Tags to simulation variables or to user-defined custom variables and select the independent/dependent variables.

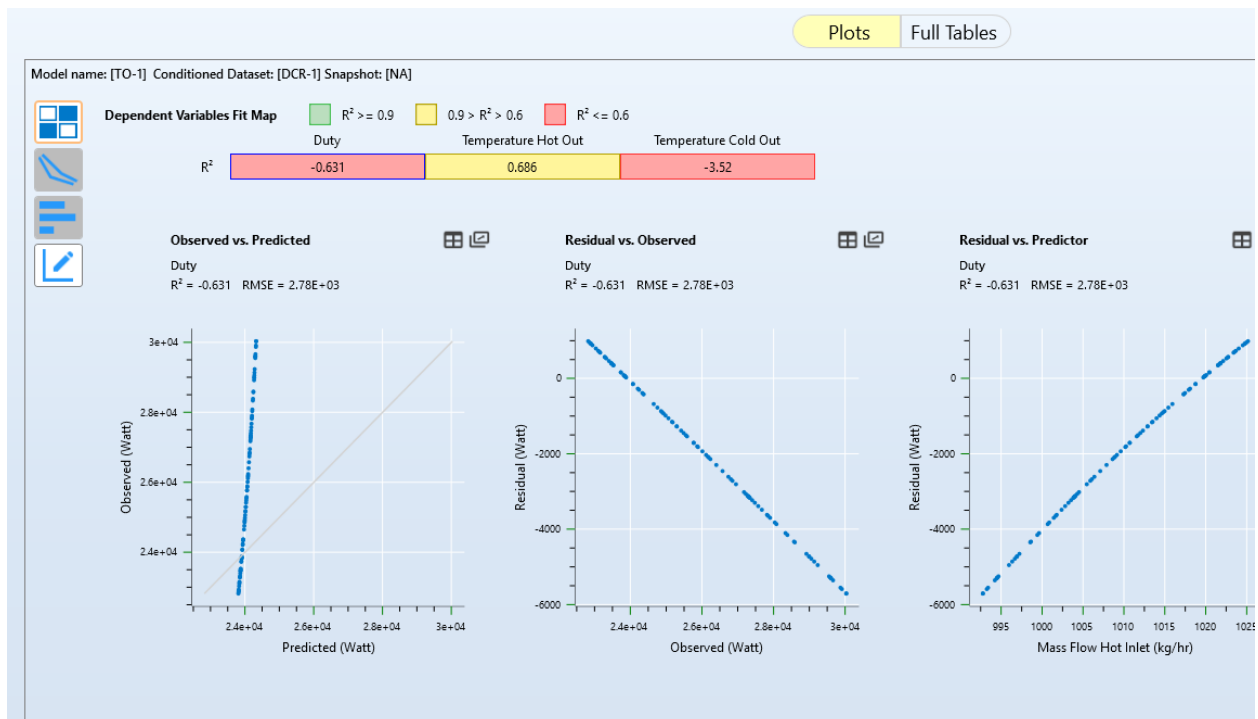
4 Map Tags to simulation variables or to user-defined custom variables
Assign each Tag as Independent or Dependent (simulation inputs = Independent, simulation outputs = Dependent)

Include in Model	Tag	Description	Units	Variable Type	Variable Name	Independent	Dependent
☒	A1001.PV	Mass Flow Hot Inlet	kg/hr	Simulation	Streams.HOTIN.Input.TOTFLOW.MIXED	☒	☐
☒	A1002.PV	Temperature Hot Feed	C	Simulation	Streams.HOTIN.Input.TEMP.MIXED	☒	☐
☒	A1003.PV	Mass Flow Cold In	kg/hr	Simulation	Streams.COLDIN.Input.TOTFLOW.MIXED	☒	☐
☒	A1004.PV	Temperature Cold Feed	C	Simulation	Streams.COLDIN.Input.TEMP.MIXED	☒	☐
☒	B1005.PV	Duty	Watt	Simulation	Blocks.HX.Output.HX_DUTY	☐	☒
☒	B1006.PV	Temperature Hot Out	C	Simulation	Streams.HOTOUT.SharedData.Substreams.IV	☐	☒
☒	B1007.PV	Temperature Cold Out	C	Simulation	Streams.COLDOUT.SharedData.Substreams.IV	☐	☒

19. Save the file as HXFouling_Mapped.apwz. A sample file has also been provided.

Note: Please map the EDR exchanger as shown in step 2 if directly using this file.

20. Click Evaluate from the ribbon.
21. Observe the results after the run.



22. The predictions for outlet temperatures and duty of the exchanger for the exchanger based on constant fouling factors are not great. We will now build a hybrid model to get better predictions for the fouling factors depending on the process conditions.

23. Save the file as HXFouling_Evaluation.apwz (solution file is provided).

Note: Please map the EDR exchanger as shown in step 2 if directly using this file.

Step 4 – Build, Train and Validate Model

24. Provide reasonable bounds to the fouling factors in this case between 0 and 100 sec-sqcm-K/cal.

3 Map simulation input variables as Neural Network Outputs

New Copy Paste

Neural Network

Trained Variable	Description	Lower Limit	Upper Limit	Initial	Units
Blocks.HX.Input.HOT_FOULING	Hot side fouling f...	0	100	4.1868	sec-sqcm-K/cal
Blocks.HX.Input.COLD_FOULING	Cold side fouling...	0	100	41.868	sec-sqcm-K/cal

25. Go to the neural network configuration and update the number of training epochs to 40.

> Neural Network Configuration

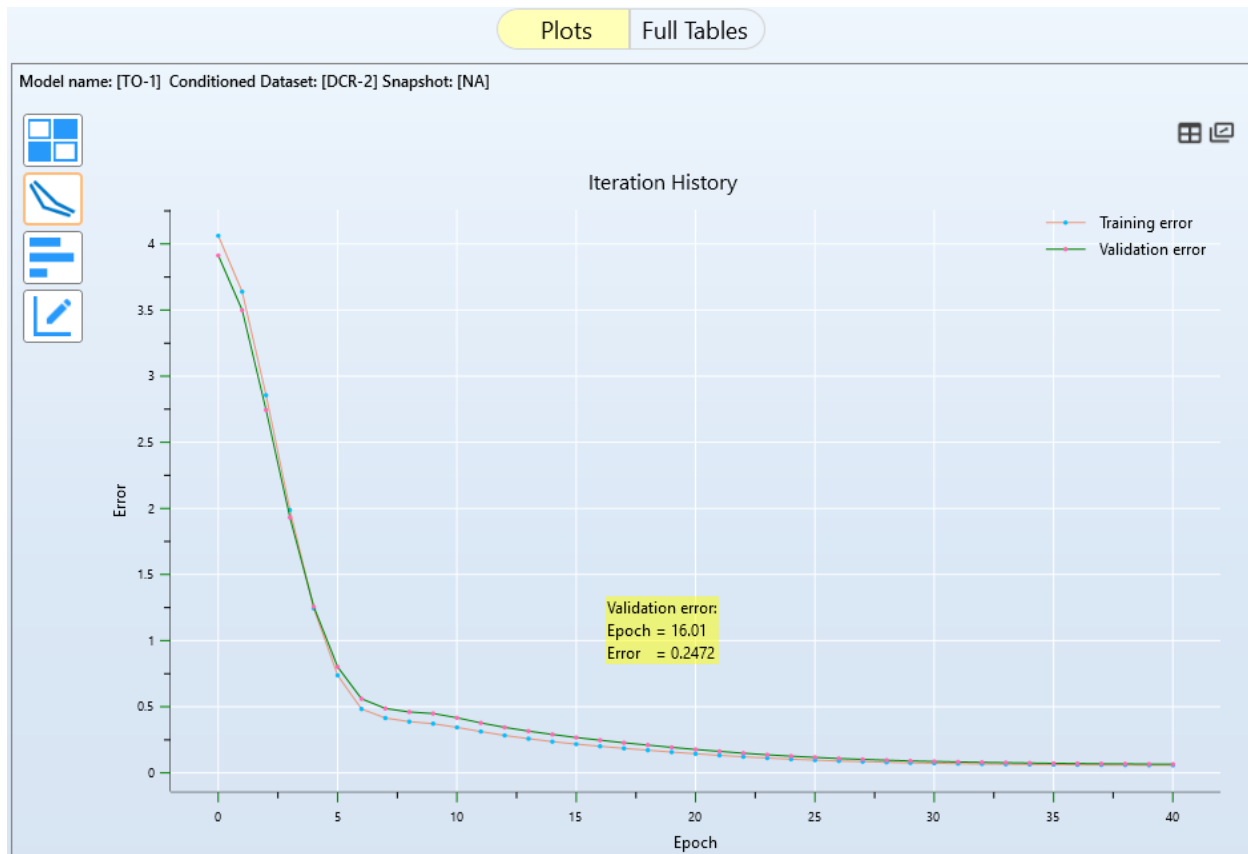
Basic

Training set fraction 0.8 Hidden Layers 1 Training Epochs 40 Learn

Hidden Layer	Neurons	Activation	Activation Parameter
1	8	Selu	0.1

Output Layer

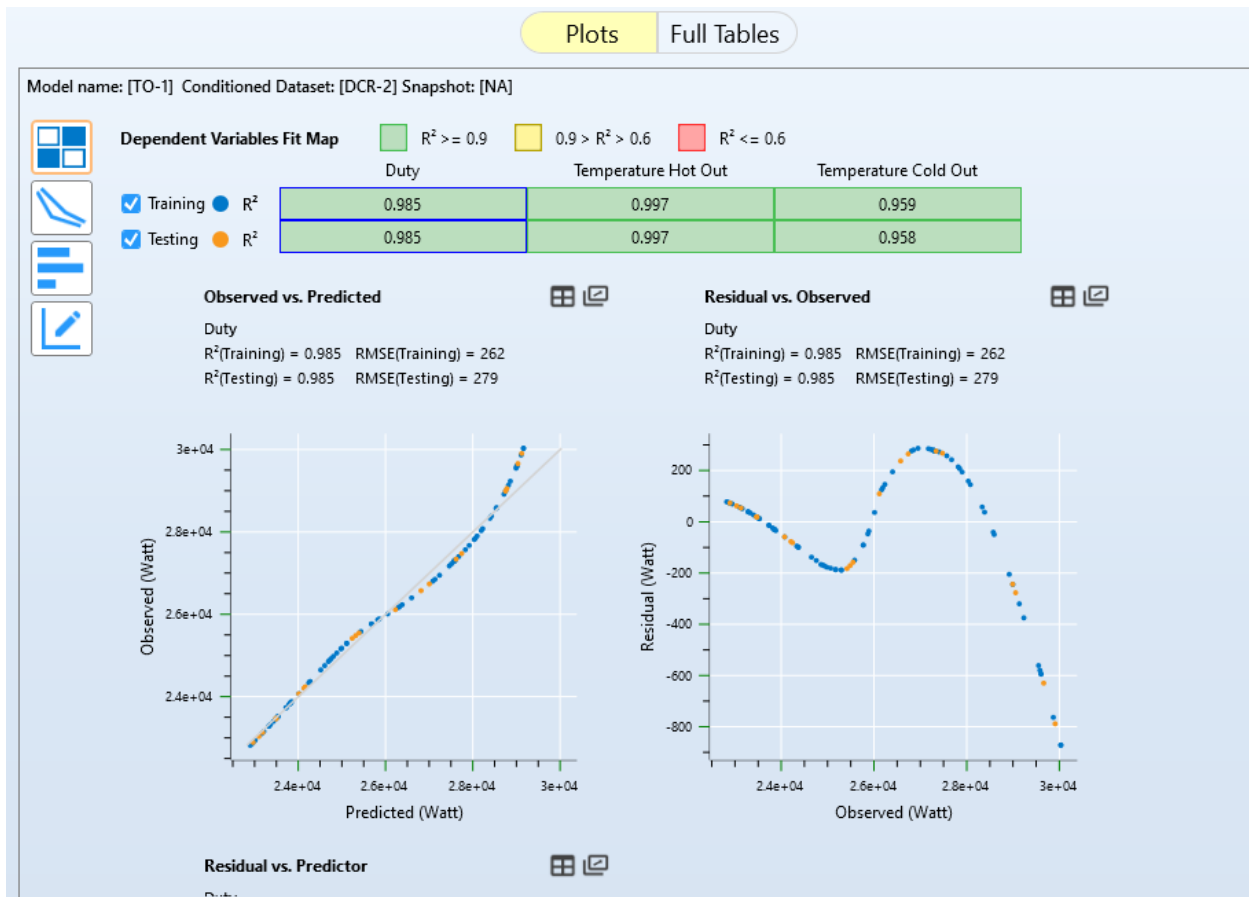
26. Click on Train to allow the neural network to train the hybrid model. It may take some time, so please be patient. The AI Training Progress will pop up and will show the training and validation error decreasing.



27. Review the results at the end of the Training run. The model should give you a good fit.

28. Save the file HXFouling_Trained.apwz (solution file is provided).

Note: Please map the EDR exchanger as shown in step 2 if directly using this file.



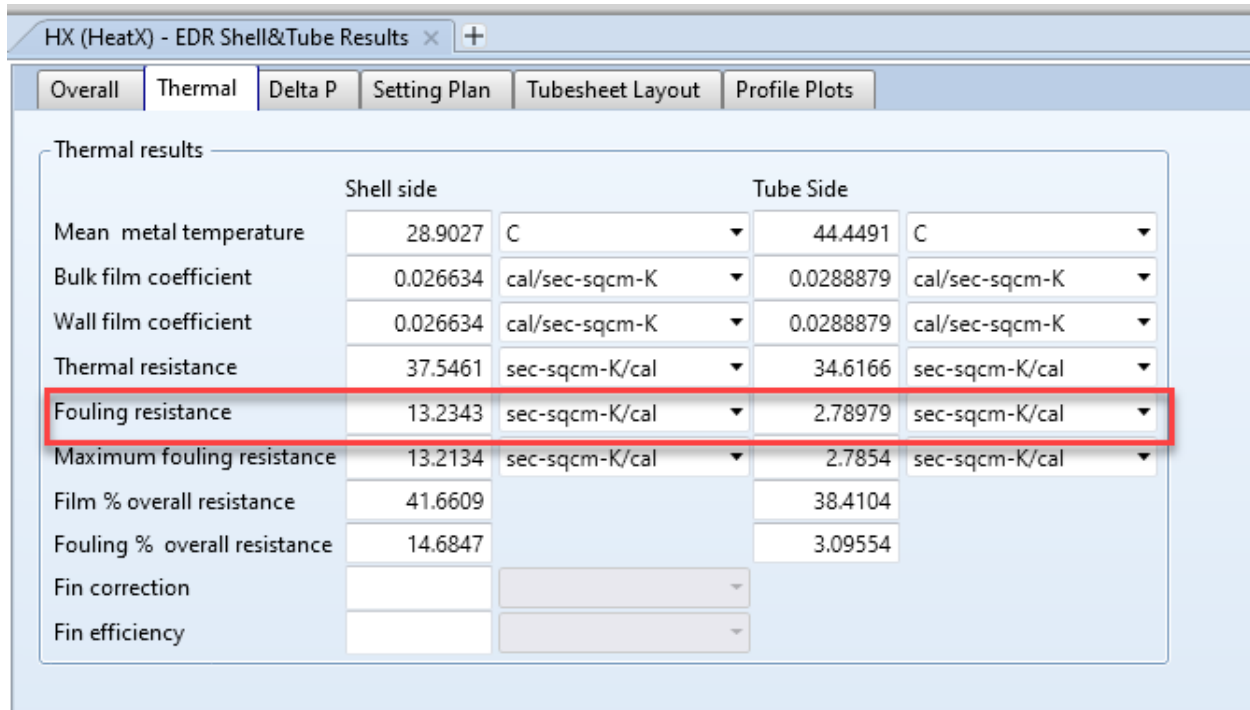
29. To take a snapshot of the current trained model, click Take Snapshot on the AI Training ribbon tab.

30. Feel free to change the neural network configuration and train the model. Save a snapshot each time.

31. Go to Manage Snapshots and select Apply Snapshot.

Step 5 – Deploy Model

32. Click on Deploy to deploy the model to the flowsheet and re-run the Aspen Plus file.
33. The model should now run with updated fouling factors based on the conditions specified. As you update the inlet flows and temperature, you will see the fouling factors update.



	Shell side		Tube Side	
Mean metal temperature	28.9027	C	44.4491	C
Bulk film coefficient	0.026634	cal/sec-sqcm-K	0.0288879	cal/sec-sqcm-K
Wall film coefficient	0.026634	cal/sec-sqcm-K	0.0288879	cal/sec-sqcm-K
Thermal resistance	37.5461	sec-sqcm-K/cal	34.6166	sec-sqcm-K/cal
Fouling resistance	13.2343	sec-sqcm-K/cal	2.78979	sec-sqcm-K/cal
Maximum fouling resistance	13.2134	sec-sqcm-K/cal	2.7854	sec-sqcm-K/cal
Film % overall resistance	41.6609		38.4104	
Fouling % overall resistance	14.6847		3.09554	
Fin correction				
Fin efficiency				

34. Save the file HXFouling_Deployed.apwz (solution file provided).

Note: Please map the EDR exchanger as shown in step 2 if directly using this file.